

Practice 4.1

1. Peter wants to repaint his 700-square-foot apartment. He calculates that he has 3500 square feet of wall space to paint. (He will not paint the floor or the ceiling.) If each gallon of paint will cover 350 square feet of wall space, how many gallons will Peter need?

A. 2
B. 5
C. 10
D. 15

$$\frac{3500 \cancel{\text{ft}^2}}{350 \cancel{\text{ft}^2} / \text{gal}} = 10 \frac{1}{1} = 10 \text{ gal}$$

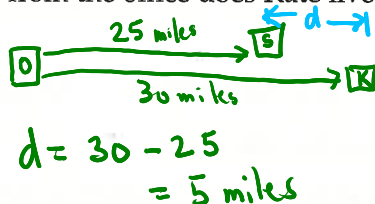
2. For a family get-together, Darryl wants to be sure that each child gets 2 party favors. The party favors cost \$3 each, and there are 11 children coming. How many party favors will Darryl need?

A. 9
B. 11
C. 18
D. 22

$$(11 \cancel{\text{children}})(2 \frac{\text{pf}}{\cancel{\text{child}}}) = 22 \text{ pf}$$

3. Sarah and Kate live 18 miles apart, and they both work at the same office. If Sarah lives 25 miles from the office and Kate lives 30 miles from the office, how many miles farther from the office does Kate live than Sarah?

A. 5
B. 7
C. 12
D. 15



$$d = 30 - 25 = 5 \text{ miles}$$

For question 4, write your answer on the line provided.

4. The Navarro family uses an average of 225 gallons of water per day, 5 gallons of which goes through the family's water filter. The Navarros' water filter can process 450 gallons before it needs to be replaced. After how many days of average water use will the family need to replace their filter?

$$\frac{450 \cancel{\text{gal}}}{5 \cancel{\text{gal}} / \text{day}} = 90 \frac{1}{1} = 90 \text{ days}$$

$$= 90 \text{ days}$$

Questions 5 and 6 are based on the following information.

Joyce owns a beauty salon, and she has posted the following information in her salon.

Service	Minutes to Complete	Price
Manicure	30	\$15
Pedicure	30	\$25
Manicure & Pedicure	45	\$35
Facial	45	\$45
Makeover	60	\$60

5. How many minutes will it take Joyce to give 3 pedicures?

A. 30
B. 55
C. 75
D. 90

$$(30 \frac{\text{min}}{\cancel{\text{ped.}}})(3 \cancel{\text{ped.}}) = 90 \text{ min}$$

6. How much more does a customer pay for the makeover than for the manicure & pedicure combination?

A. \$15
B. \$20
C. \$25
D. \$35

$$60 - 35 = 25$$

For questions 7 and 8, write your answers on the lines provided.

7. Brandon is planning his part of the local community garden. He has calculated that he can plant 6 seedlings per row, and the garden allots 7 rows to each gardener. What is the maximum number of seedlings Brandon could plant?

$$(7 \cancel{\text{rows}})(6 \frac{\text{seed}}{\cancel{\text{row}}}) = 42 \text{ seedlings}$$

8. Sumayyah knits gloves and hats each winter and sells them through a local shop. If Sumayyah received \$189 from the sale of her hats and gloves this past winter, and she spent \$27 on yarn, how much money did she earn from her knitting?

$$\text{Profit} = 189 - 27 = \$162$$

$$\text{Profit} = \text{Revenue} - \text{cost}$$

Answers and explanations begin on page 661.

Multi-Step Word Problems

The word problems on the previous two pages involved only one operation to solve. However, on the GED® *Mathematical Reasoning Test* and other tests, you may need to do several math processes to solve a problem. These are called **multi-step problems**.

When you are working on multi-step problems, there are two important ideas to keep in mind. *What is the question asking me to find?* and *Did I answer the question?*

Review this example of a multi-step problem and the problem-solving process below.

Example 1: To win a prize, Sarah's daughter has to sell 75 boxes of cookies in 3 days. If she sold 16 boxes on day 1 and 34 boxes on day 2, how many boxes would she need to sell on day 3 to win the prize?

- (1) 18
- (2) 25
- (3) 50

Step 1. What is the question asking me to find?

The question asks me to find how many boxes Sarah's daughter needs to sell on the third day to make her goal of 75 boxes.

Step 2. What information from the problem do I need?

I know the total number of boxes she needs to sell (75 boxes).

I know how many boxes she sold on day 1 (16 boxes) and day 2 (34 boxes).

Step 3. Do I need to do only one operation, or do I need to do more than one?

I can add to find how many boxes Sarah's daughter has sold so far: $16 + 34 = 50$.

However, that doesn't tell me how many boxes she needs to sell on day 3. I need one more step.

Step 4. What is my solution?

$75 \text{ boxes total} - 50 \text{ boxes sold} = 25 \text{ boxes for day 3}$. The correct answer is (2) 25.

Step 5. Does my answer make sense? Did I answer the question?

By taking the additional step, I answered the question of how many boxes need to be sold on day 3.

This sample multi-step problem involved only two operations: adding and subtracting. On the actual test, you may need to do three or more operations to answer a question, so read and think carefully about each problem.

As you think about the example, note the most common mistake people make with multi-step problems. Often they stop short of answering the question. Notice that choice (3) 50 is a **partial solution**, but it is not the answer to the question. Using the five-step problem-solving process will help you avoid that mistake.

Practice 4.2

Questions 1–3 are based on the information below.

Farhana's produce company distributes to several restaurants. The table below shows how many cases of different produce each restaurant ordered from Farhana's company in July.

Produce Orders in July					
	Asparagus	Boston Lettuce	Carrots	Romaine Lettuce	Tomatoes
Restaurant A	2	3	1	4	3
Restaurant B	4	4	2	2	1
Restaurant C	0	0	3	4	3
Restaurant D	1	2	2	3	4
Restaurant E	3	0	3	2	1

1. If Boston lettuce costs \$17 per case and romaine lettuce costs \$23 per case, how much did Restaurant D spend on lettuce ordered from Farhana in July?

A. \$85
B. \$93
☒ C. \$103
D. \$143

Boston: $(2)(17) = \$34$
Romaine: $(3)(23) = \$69$
total = \$103

2. Delivery costs \$2 per case for the first 5 cases and \$1 per case for each additional case of produce. What was Restaurant B's delivery charge in July?

A. \$13
☒ B. \$18
C. \$20
D. \$26

total orders = 13
1st 5: cost = $(\$2)(5) = \10
others: $(\$1)(8) = \8
total cost = \$18

3. If asparagus costs \$22 per case and tomatoes cost \$15 per case, which of the following restaurants spent the most on asparagus and tomatoes combined?

A. Restaurant A
☒ B. Restaurant B
C. Restaurant C
D. Restaurant D

A: $2(22) + 3(15) = 44 + 45 = \89

B: $4(22) + 1(15) = 88 + 15 = \103

C: $3(15) = \$45$

D: $1(22) + 4(15) = 22 + 60 = \82

4. At a certain store, loose-leaf paper comes only in packages of 400 sheets. If a student buys enough paper at this store to fill 3 binders with 150 sheets of paper each, how many sheets will be left over?

A. 17
B. 50
☒ C. 350
D. 450

sheets used = $(3 \text{ binders})(150 \frac{\text{sheets}}{\text{binder}}) = 450 \text{ sheets}$
 $(2 \text{ pack.})(400 \frac{\text{sheets}}{\text{pack}}) = 800 \text{ sheets}$
 $800 \text{ sheets} - 450 \text{ sheets} = 350 \text{ sheets}$

5. Three friends are baking cupcakes for a bake sale. Each batch of 24 cupcakes requires 2 cups of flour. The friends have a single 5-pound bag of flour that contains 19 cups of flour. How many whole batches of cupcakes can they bake?

☒ A. 9
B. 38
C. 216
D. 228

$\frac{19 \text{ cups}}{2 \frac{\text{cups}}{\text{batch}}} = 9.5 \text{ batches}$

6. A certain health insurance plan costs \$3000 per year for a family of six. If each member of the family has \$750 in medical expenses in a year, and the plan pays 100 percent of those expenses, how much will the family save by purchasing the plan?

☒ A. \$1500
B. \$2000
C. \$2500
D. \$3000

total expense = $(6 \text{ people})(750 \frac{\text{dollars}}{\text{person}}) = \4500
so $\$4500 - \$3000 = \$1500$

Answers and explanations begin on page 661.

Lesson 5: Distance and Cost

Key Ideas

- The following all indicate multiplication:

$$n \times r \quad nr \quad n(r) \quad n \cdot r$$

- The following indicate division:

$$c \div n \quad \frac{c}{n}$$

- A formula can be rewritten to solve for each of its variables.

$$c = nr \quad \frac{c}{n} = r \quad \frac{c}{r} = n$$

$$d = rt \quad \frac{d}{r} = t \quad \frac{d}{t} = r$$

ON THE GED® TEST

Substitute all of the values for variables back into a formula to check your answer.

Distance

On the *Mathematical Reasoning Test*, you'll see questions that will require you to apply formulas. When a relationship is constant, use a formula to set up the information about how the different parts relate to each other. One of the most commonly tested formulas deals with distance, rate, and time. Distance is a product of the rate at which something travels and the amount of travel time:

$$\text{Distance} = \text{rate} \times \text{time}, \quad \text{or} \quad d = rt$$

Notice that letters, or variables, can be used to represent the different parts of a formula. A formula allows you to substitute known values for certain variables and solve for the unknown variable.

Example 1: How many miles can you travel if you drive at an average speed of 55 miles per hour for 3 hours?

- You know the rate (55 miles per hour) and the time (3 hours). Substitute the values in the distance formula. $\text{distance} = 55 \times 3$

- Multiply to find the distance. $55 \times 3 = 165 \text{ miles}$

Cost

Another helpful formula is the cost formula. It expresses the relationship between cost, the number of units, and rate (price per unit). **NOTE:** The word *per* means for every one unit. The cost formula can be written as follows:

$$\text{Total cost} = (\text{number of units}) \times (\text{price per unit}), \quad \text{or} \quad c = nr$$

Example 2: At a bakery, a package of frosted cookies is priced at \$3 per package. If a teacher treats her class by buying 4 packages, how much would the cookies cost before tax?

- You know the number of units (4 packages) and the price per unit (\$3 per package). Substitute the values in the cost formula. $c = nr$
 $\text{total cost} = 4 \times \3

- Multiply to find the total cost. $4 \times \$3 = \12

If you know any two of the three variables in a formula, you can solve for the third variable.

Example 3: Max bought a set of 4 floor mats for \$44. How much was the price per floor mat?

- You know the total cost (\$44) and the number of units (4 floor mats). Rewrite the formula to solve for the price per unit (r). $c = nr$
 $r = \frac{c}{n}$
- Substitute the known values in the cost formula. Divide to find the price per unit. $\frac{\$44}{4} = \11

Practice 5

- A. Each problem below includes two of the three variables from either the distance formula or the cost formula. Write the missing variable you need to solve for. Then decide which of the following formula variations you would use in each situation. The first one is done for you.

$$d = rt$$

$$\frac{d}{r} = t$$

→ speed

$$\frac{d}{t} = r$$

$$c = nr$$

$$\frac{c}{n} = r$$

↑ Cost per item

$$\frac{c}{r} = n$$

- | | |
|--|--|
| <p>1. Given: distance and time
Solve for: rate
Formula: $\frac{d}{t} = r$</p> <p>2. Given: rate and time
Solve for: _____
Formula: _____</p> <p>3. Given: distance and rate
Solve for: _____
Formula: _____</p> | <p>4. Given: cost and number of units
Solve for: _____
Formula: _____</p> <p>5. Given: number of units and price per unit
Solve for: _____
Formula: _____</p> <p>6. Given: cost and price per unit
Solve for: _____
Formula: _____</p> |
|--|--|

- B. Use the formulas provided in part A above to help you set up the problems. Solve for the unknown variable.

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| <p>7. Find the total cost of 4 flats of plants at \$12 each.</p> <p>8. Find the total cost of 12 boxes of cookies if each box costs \$3.</p> <p>9. If 4 tires cost \$320, how much does a single tire cost?</p> <p>10. How many tickets would you get for \$25 if raffle tickets cost \$5 apiece?</p> <p>11. If you paid \$20 for 10 bus transfer tickets, how much did you pay per ticket?</p> | <p>12. Find the distance traveled by a car averaging 60 miles per hour for 3 hours.</p> <p>13. Find the distance traveled by a train averaging 50 miles per hour for 4 hours.</p> <p>14. How long does it take for a bus to travel 25 miles at an average rate of 25 miles per hour?</p> <p>15. If a train travels 270 miles in 3 hours, what is the train's average speed?</p> <p>16. How long does it take to complete a delivery route of 75 miles at a rate of 25 miles per hour?</p> |
|---|---|

9. $c = nr$
 $\frac{320}{4} = \frac{nr}{4}$
 $r = \$80 \text{ per tire}$

15. $d = rt$
 $\frac{270}{3} = \frac{r(3)}{3}$
 $90 \frac{\text{miles}}{\text{h}} = r$

- C. Choose the one best answer to each question.

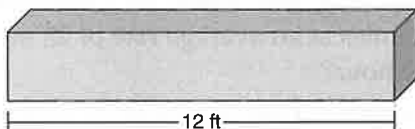
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| <p>17. A company sold a total of \$640 in gift boxes. If the gift boxes cost \$20 apiece, how many gift boxes did the company sell?</p> <p>A. 32
B. 320
C. 660
D. 1280</p> | <p>18. A truck driver traveled 275 miles in 5 hours. What was his average speed in miles per hour?</p> <p>A. 1375
B. 280
C. 270
D. 55</p> |
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Answers and explanations begin on page 661.

Number Sense and Problem Solving Practice Questions

Directions: You MAY use your calculator.

1. A beverage container holds 12 servings. If the serving size is 8 ounces, how many ounces does the container hold in all?
A. 20
B. 32
C. 48
D. 96
2. Sales at 3 concession stands are \$839, \$527, and \$726. What is the total amount in sales?
A. \$1581
B. \$2092
C. \$2178
D. \$2517
3. If you want to cut 24 two-foot braces, how many boards of the length shown below would you need?
4. Using your calculator, find the value of $\sqrt{441}$.
A. 11
B. 21
C. 221
D. 441
5. Angelo bought a used car with 39,451 miles on it. If the car now has 70,040 miles on it, how many miles has Angelo driven the car?
A. 30,589
B. 39,459
C. 70,040
D. 109,491
6. Use your calculator to solve this problem. If Emory paid 20 percent of \$3280 as a down payment, how much was the down payment?
A. \$164
B. \$328
C. \$656
D. \$6560



- A. 4
- B. 6
- C. 12
- D. 24

7. Inventory shows that a warehouse has 45 printers in stock. If each printer is valued at \$125, what is the total value of the printer inventory?

- A. \$5625
- B. \$170
- C. \$80
- D. \$45

$$\text{total} = (45 \text{ printers})(125 \text{ dollars per printer}) = \$5625$$

8. Lydia can drive 180 miles in 3 hours. On Tuesday, she drove for 7 hours at that rate. How many miles did she drive on Tuesday?

- A. 60
- B. 420
- C. 600
- D. 1260

9. Janelle wants to drive from Danville to Brownsville. If she averages 60 miles per hour, how many hours will it take her to drive the distance?



- A. 6
B. 8
C. 60
D. 540
10. A company sold a total of \$1440 in gift bears for Valentine's Day. If the gift bears cost \$15 apiece, how many gift bears did it sell?
- A. 15
B. 96
C. 144
D. 1440
11. In addition to interest charges, Richard's credit card company charges a \$25 late fee for payments made after the payment due date. If he was charged a late fee for 8 monthly bills, how much could he have saved by paying the bills on time?
- A. \$200
B. \$80
C. \$33
D. \$25
12. A waiter has seven \$5 bills and eighteen \$1 bills from tips. In all, how much does he have in tips?
- A. \$18
B. \$25
C. \$35
D. \$53

13. April has taken her car in for the recommended oil and filter change every 3,500 miles. If April bought her car brand-new, and the odometer now shows just over 17,500 miles, how many oil changes has her car received?

- A. 5
B. 14
C. 123
D. 1236

$$\frac{(17500 \text{ miles})}{3500 \text{ miles oil change}} = 5 \text{ oil changes}$$

14. A clinic treated 536 children over a 4-month period. At this rate, how many children did the clinic treat in 1 month?

- A. 134
B. 536
C. 540
D. 2144

15. Attendance at a local play was 438 Friday night, 820 Saturday night, and 636 Sunday afternoon. How many more people attended the play on Sunday than on Friday?

Write your answer on the line below.

16. Raquel has 4 payments left on her car. If each payment is \$268, how much does she still owe on her car?

Write your answer on the line below.

\$1072

$$(4 \text{ payments}) (268 \text{ dollars payment})$$

$$= \$1072 \text{ dollars}$$

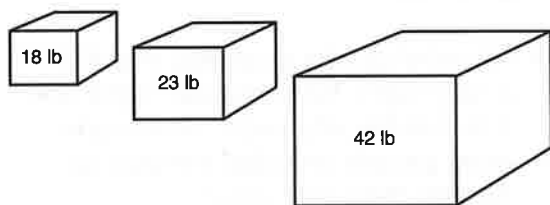
17. In what order should items weighing 51 pounds, 40 pounds, 48 pounds, and 44 pounds be stacked if you want them in order from heaviest to lightest?

A. 51, 44, 40, 48
B. 40, 44, 48, 51
C. 51, 48, 44, 40
D. 51, 40, 44, 48

18. Which of the following correctly shows 2,354,769 rounded to the nearest ten thousand?

A. 2,400,000
B. 2,355,000
C. 2,350,000
D. 2,000,000

19. What is the total weight, in pounds, of the packages below?



- A. 83
B. 65
C. 60
D. 42
20. Jason paid a \$14 dinner bill with a \$20 bill. How much change should he receive?
- A. \$6
B. \$7
C. \$14
D. \$34

21. Maria spent 8 minutes installing a new showerhead, 33 minutes rodding out a drain, and 18 minutes fixing a leaking faucet. About how many minutes did it take Maria to complete all three tasks?

A. 90
B. 60
C. 30
D. 20

22. Carla drove 248 miles in 4 hours. What was her average rate of speed, in miles per hour?

Write your answer on the line below.

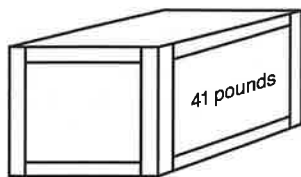
23. A bulk bag of nuts contains 144 ounces of nuts. If the nuts are packaged in smaller 8-ounce bags, how many bags will there be?

A. 8
B. 12
C. 18
D. 136

24. If you drove 299 miles on 9 gallons of gasoline, about how many miles per gallon did the car get?

A. 10
B. 30
C. 270
D. 300

25. A shipment of 33 crates like the one shown below is delivered. Approximately how many pounds did workers unload?



- A. 40
B. 120
C. 1200
D. 2000
26. Four friends bought a birthday cake for \$21 and balloons for \$15. If they divided the cost equally, how much did each friend pay toward the birthday party?
- A. \$4
B. \$5
C. \$9
D. \$36
27. David paid \$20 toward a dinner bill of \$128. If the remainder of the bill is divided equally among the remaining 9 people in the group, how much should each person other than David pay?
- A. \$12
B. \$18
C. \$20
D. \$108
28. A driver traveled 4 hours at an average rate of 65 miles per hour. How many miles did the person drive?
- A. 61
B. 69
C. 240
D. 260

29. Using the following information, how much would a large pizza with 3 toppings cost?

Large 1-Topping Pizza for \$14 \$2 for Each Additional Topping

- A. \$20
B. \$18
C. \$16
D. \$14
30. Bagels are 2 for \$1. What is the maximum number of bagels you could buy for \$7?
- A. 7
B. 10
C. 14
D. 15
31. How many months would it take to save \$1800 at \$75 per month?
- Write your answer on the line below.
- _____
32. How many 45-page documents would a binder hold if its maximum capacity is 630 sheets of paper?
- Write your answer on the line below.
- _____

Answers and explanations begin on page 662.

Decimals and Fractions

Lesson 1: Decimal Basics

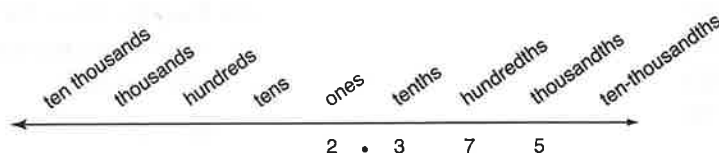
Key Ideas

- The decimal point separates the whole number from the decimal part.
- To round decimals, use the same rules as with rounding whole numbers.
- Adding zeros to the right of the last digit after the decimal does not change the number's value.

The Decimal System

Decimals are numbers that use place value to show amounts less than 1. You already use decimals when working with money. For example, in the amount \$10.25, you know that the digits to the right of the **decimal point** represent cents, or hundredths of a dollar.

The first four decimal place values are labeled on the chart below.



The number 2.375 is shown on the chart. Read *and* in place of the decimal point. After reading the decimal part, say the place value of the last decimal digit. This number would be read "two *and* three hundred seventy-five *thousandths*."

Rounding

Round decimals the same way you round whole numbers.

Example 1: A calculator display reads 3.62835. Round to the hundredths place.

1. Look at the digit to the right of the hundredths place. 3.62835
2. If the digit to the right is 5 or greater, round up. If the digit is less than 5, don't change the number. Then drop all digits to the right of the place you are rounding to. Since 8 is greater than 5, round up.
3.62835 rounds to
3.63.

Comparing and Ordering

Comparing decimals is similar to comparing whole numbers.

Example 2: Matt ran the 400-meter race in 45.8 seconds. Alonzo ran the same race in 45.66 seconds. Which runner had the faster time?

1. Line up the decimal points. Add a zero at the **end** of 45.8 so that both times have the same number of digits after the decimal. 45.80
45.66
2. Compare the decimal parts of the numbers as though they were whole numbers. **Alonzo's time was faster.** 80 is greater than 66, so 45.8 is greater than 45.66.

ON THE GED® TEST

The Mathematical Reasoning Test may show a zero written before the decimal point. The zero does not change the value of the number.
Example: 0.75 = .75

When you compare more than two numbers, it is helpful to compare one place-value column at a time, working from left to right.

Example 3: Arrange the numbers 0.85, 1.8, 0.8, and 0.819 in order from greatest to least.

- | | |
|---|---------------------------------------|
| 1. Write the numbers in a column, lining up the decimal points.
Add zeros so that the numbers have the same number of decimal places. | 0.850
1.800
0.800
0.819 |
| 2. Compare the digits, working from left to right. Only 1.8 has a whole number part, so it is greatest. The remaining numbers each have 8 in the tenths column. Looking at the hundredths column, 0.85 is next, followed by 0.819. The least number is 0.8. | In order: 1.8
0.85
0.819
0.8 |

Practice 1

A. Round these numbers as directed.

- | | |
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| 1. Round 3.75 to the tenths place. | 4. Round 0.66667 to the nearest thousandth. |
| 2. Round 5.908 to the ones place. | 5. Round 8.125 to the nearest tenth. |
| 3. A calculator display reads 0.4285714.
Round to the nearest hundredth. | 6. A calculator display reads 2.7142857.
Round to the nearest thousandth. |

B. In each of the following pairs, which number is greater?

- | | | |
|------------------|------------------|---------------------|
| 7. 0.45 or 0.449 | 9. 4.68 or 4.086 | 11. 1.0275 or 1.029 |
| 8. 0.008 or 0.08 | 10. 0.75 or 1.85 | 12. 0.14 or 0.104 |

C. Write these numbers in order from least to greatest.

- | | |
|---------------------------|----------------------------|
| 13. 5.6 5.08 5.8 5.802 | 15. 14.005 4.52 4.8 4.667 |
| 14. 0.1136 0.12 0.2 0.115 | 16. 0.8023 0.8 0.803 0.823 |

D. Choose the one best answer to each question.

- | | |
|---|---|
| 17. In a circuit board assembly, the weights of three parts are 0.572 grams, 0.0785 grams, and 0.6 grams. Which of the following lists the weights in order from <u>greatest to least</u> ? | 18. Which of the following correctly shows 1.3815 rounded to the nearest hundredth? |
| A. 0.0785 g, 0.572 g, 0.6 g | A. 1.4 |
| B. 0.6 g, 0.0785 g, 0.572 g | B. 1.382 |
| C. 0.6 g, 0.572 g, 0.0785 g | C. 1.381 |
| D. 0.572 g, 0.6 g, 0.0785 g | D. 1.38 |

Answers and explanations begin on page 662.